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Received: 15 August 2025

Accepted: 17 November 2025

Published online: 29 November 2025

Cite this article as: Tran C.D., Kuchar M., Sotola V. *et al.* Sensor fault diagnosis strategy based on rotor flux observers in three-phase induction motor drive. *Sci Rep* (2025). <https://doi.org/10.1038/s41598-025-29381-9>

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Sensor Fault Diagnosis Strategy based on Rotor Flux Observers in Three-phase Induction Motor Drive

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ABSTRACT

Sensors are crucial in induction motor drives (IMDs) by providing real-time feedback essential for precise and stable speed control. Malfunctions in speed or current sensors can seriously impair system performance and reduce reliability. This study proposes a fault-tolerant control (FTC) strategy that combines a rotor flux observer with a signal similarity-based diagnosis algorithm to detect, isolate, and compensate for sensor faults within the field-oriented control (FOC) framework. Once a fault is detected, the control system is reconfigured instantly by replacing faulty measurements with estimated values, ensuring stability during operation. The effectiveness of the method is evaluated through both simulation and experimental tests on a laboratory platform using a TMS320F28335 digital signal controller. Simulation results confirm the approach's ability to identify different fault types with minimal delay accurately. Real-time experiments further validate the method's practical application under typical operating conditions. Across all test scenarios, the system successfully maintains performance by smoothly switching between sensor-based and sensorless control modes. These results highlight the robustness and dependability of the proposed FTC scheme for IMDs experiencing sensor faults.

Introduction

The three-phase induction motor, with its outstanding advantages of a simple structure, strong load capacity, straightforward starting, stable operation, and high performance, is one of the most popular and vital types of motors applied in various industrial fields^{1,2}. Along with the strong development of the power electronics field, especially the emergence of inverter technology, the ability to flexibly and accurately control the speed of three-phase induction motors has become a core tool in various industries, including the machinery manufacturing industry, textile industry, and automobile industry, etc.^{3,4}. Motor speed control methods are typically divided into three main groups, including scalar control (V/F), field-oriented control (FOC), and direct torque control (DTC). The V/F strategy, with a straightforward structure, is suitable for open-loop control, while FOC and DTC, based on dynamic models, are ideal for closed-loop control^{5,6}. In the industry, the FOC method, with its control feature similar to that of the DC motor, is preferred in drive systems for applications that require high control accuracy^{7,8}. However, a control system applying the FOC technique also needs a proper hardware drive structure with full sensors to control motor speed and achieve demand performance^{9,10}. The sensors integrated into the drive system provide the controller with the instantaneous status of the motor, enabling it to perform the appropriate control algorithm; therefore, the accuracy of the sensors plays a significant role in ensuring the reliability and stability of the motor drive. Thus, determining the quality of the feedback signals from the sensors is a core requirement in ensuring the effectiveness of the motor speed control method^{11,12}.

In the motor drive system, three types of sensors are used: DC link voltage sensors, current sensors, and speed sensors. However, the accuracy of estimators and observers can be reduced due to the influence of drift in voltage measurement. Therefore, the voltage sensor

is not used in many applications but is replaced by a constant value combined with the inverter switching signal¹³. Hence, this research focuses solely on the fault diagnosis method of sensors, specifically stator current and motor speed. To increase the reliability and stable operation of motor drives in industrial applications, drive controllers must incorporate a fault-tolerant control (FTC) function. Recently, the FTC technique has garnered considerable attention from researchers; various FTC methods have been proposed for integration into motor controllers, yielding impressive achievement results. FTC methods can be divided into two basic groups: Passive-FTC and Active-FTC, each with advantages and disadvantages. The group of methods belonging to Passive-FTC is implemented from the design stage, utilizing redundant hardware structures to maintain stable operation with predefined, simple types of faults^{14,15}. In contrast, the Active-FTC method group has a wider operating range, corresponding to various sensor faults, and utilizes multiple diagnostic algorithms. A typical Active-FTC method includes three main steps: sensor fault detection, fault signal isolation, and control mode reconstruction.

To comprehensively address the sensor signal failure problem, this research focuses on the group of Active-FTC methods, with the objective of using FOC platform techniques to control speed and torque in an induction motor drive. The distinction between Active-FTC methods primarily relies on the fault detection technique used for sensor signals.

Sensor fault diagnosis (SFD) approaches can be generally categorized based on their foundational methods. The first category employs observers or estimators for comparison purposes to assess the agreement with a predefined signal. In this category, techniques like the Extended Kalman Filter (EKF), the Luenberger Observer and Sliding Mode Observer (SMO)¹⁶⁻²³ are well known and have proven effective in addressing faults in current and speed sensors, particularly at low-speed conditions. The second category consists of methods that inspect anomalies in the measured signals directly, often using difference operators or derivative-based indices²⁴⁻²⁸. Lastly, there are more complex methods that utilize specialized fault identification mechanisms, such as Dulmage-Mendelsohn decomposition or asymmetry indices, to improve resilience across various operating conditions^{29,30}. The representative FTC and SFD approaches from prior literature are summarized in Table 1, highlighting their applicability to current and speed sensor fault scenarios.

FTC approach	Fault detection method	Against current sensor fault	Against speed sensor fault
FTC methods apply observers (or estimators)			
Extended Kalman Filter (EKF)	Residual comparison ¹⁶	Yes	Yes
	State observation ¹⁷	No	Yes
	Sensorless speed estimation ¹⁸	No	Yes
Adaptive stator flux observer	Fault detection and isolation (FDI) logic, speed error threshold ¹⁹	No	Yes
Luenberger observer	Residual comparison and FDI logic ²⁰	Yes	No
	Speed estimation from voltage and current ²¹	No	Yes
Current space vector estimator	Three vector residual comparison ²²	Yes	Yes
Sliding Mode Observer (SMO) combining three current Observer based on Kirchoff's current law	Residual comparison ²³	Yes	Yes
SFD methods check for anomalies in the signal itself			

Third-difference operator (TDO)	TDO detection and flux observer ²⁴	Yes	No
	TDO operator ²⁶	Yes	No
	TDO speed and residual comparison ²⁷	Yes	No
Third-difference operator (IMD without speed sensor)	TDO detection and EKF residual ²⁵	Yes	No
Third-difference operator combining a single discrete derivative operator	TDO high speed, single discrete derivative operator low speed ²⁸	Yes	No
SFD methods use a complex fault identification function			
The Dulmage-Mendelsohn (DM) decomposition	DM decomposition, residual generation ²⁹	Yes	Yes
Asymmetry index combining the average function of the reference current component	Asymmetry index and dq orthogonality check ³⁰	Yes	Yes

Table 1. Summary of FTC methods in the prior literature.

After detecting the fault in the sensor type, it is necessary to accurately locate which sensor is faulty (A-phase current sensor, B-phase current sensor, or speed sensor). Next, to ensure the stable operation of the motor drive, the faulty signals are isolated from the control system (the entire current signals or motor speed signal). Finally, the control algorithm restructuring step is performed to switch from the full-sensor FOC mode to the speed-sensorless or current sensorless mode. When a sensor fault occurs, the faulty signal will be isolated immediately. In case of a speed sensor fault, the speed estimation algorithms are used to provide a feedback signal to the FOC controller³¹⁻³⁴; if a current sensor fault occurs, the estimated currents are provided accordingly^{35,36}. This study presents a unified diagnostic framework utilizing a rotor flux observer with both current and voltage models, enabling precise fault detection and localization. It supports real-time implementation and reconfiguration of control. The proposed sensor fault diagnosis method has the following distinctive features:

- Real-time integration: This method is integrated into the digital signal controller and ensures sensor FTC operation in real-time.
- Distinctive features of the diagnostic method: separates the sensor fault diagnosis process into two stages: Stage 1: determine whether a sensor fault occurs; Stage 2: locate the faulty sensor.
- Elimination of false sensor fault detection in transient states: The proposed diagnosis method limits false alarms caused by strong current fluctuations when the reference speed is suddenly changed (see Fig. 29). This is a common problem when applying the traditional comparison between the measured current and the estimated current of current observers.
- Precise localization: The method accurately identifies faulty sensors (phase A current, phase B current, or speed), allowing for targeted control adjustments. Only two current sensors are needed for the correct operation.
- Effective simplicity: The approach is based on a mathematical model of the induction motor, emphasizing simplicity, low computational demands, and ease of industrial application, thereby ensuring high diagnostic accuracy and stability under fault conditions.

The paper presents a simple yet efficient approach to fault diagnosis of the induction motor drives. The proposed sensor fault diagnosis method divides the sensor fault diagnosis process into two stages: Stage 1: Using the rotor flux signal from the two voltage models and the current model as the fundamental to detect and determine whether a sensor fault occurs; Stage 2: After determining that a sensor fault has occurred, the faulty sensor will be located according to a suitable diagnostic process. Since the rotor flux quantity from two Voltage and Current models is selected as the fundamental for sensor fault diagnosis, the proposed method is suitable for motor drive structures based on the FOC technique (which often keeps the rotor flux quantity at the rated value during various operating conditions), and especially ideal for backup sensorless mode using the MRAS group of techniques. The paper includes sections: introduction, dynamic model of the induction motor, field-oriented control, fault-tolerant control method, simulation and experimental verification of the proposed algorithm, and conclusion.

Dynamic model of the three-phase induction motor

The three-phase induction motor receives electrical energy through an electromagnetic process, which is then converted into mechanical energy at the motor shaft terminal for speed control applications. The dynamic model of a three-phase induction motor is characterized by differential equations with the state variables corresponding to the rotor flux and the stator current. The supply voltage to the stator windings is considered the input signal of the differential equation system. The differential equation system of the induction motor model is described as equations (1) and (2).

$$\mathbf{u}_s^t = R_s \mathbf{i}_s^t + s L_s \frac{d\mathbf{i}_s^t}{dt} + \frac{L_m}{L_r} \frac{d\mathbf{y}_r^t}{dt} \quad (1)$$

$$0 = -\frac{L_m}{T_r} \mathbf{i}_s^t + \left(\frac{1}{T_r} - j\omega_r\right) \mathbf{y}_r^t + \frac{d\mathbf{y}_r^t}{dt} \quad (2)$$

where: $\mathbf{u}_s^t$: Voltage in vector form; $\mathbf{i}_s^t$: Stator current in vector form; $\mathbf{y}_r^t$: Rotor flux in vector form; R_s, R_r, L_s, L_r, L_m : Stator resistance, rotor resistance, stator inductance, rotor inductance, magnetizing inductance; $T_r = \frac{L_r}{R_r}$: Rotor time; $s = \frac{L_s L_r - L_m^2}{L_s L_r}$: Leakage inductance; $\omega_r = \rho \omega_m$:

Rotor speed; ω_m : Mechanical speed; ρ : Number of pole pairs.

By applying Clarke and Park transformations, the stator current and rotor flux equations are obtained in the $[\alpha-\beta]$ reference frame, as expressed in (3) and (4).

$$\begin{cases} \frac{di_{sa}}{dt} = -Q_1 i_{sa} + Q_2 y_{ra} + Q_3 \omega_r y_{rb} + Q_4 u_{sa} \\ \frac{di_{sb}}{dt} = -Q_1 i_{sb} + Q_2 y_{rb} - Q_3 \omega_r y_{ra} + Q_4 u_{sb} \end{cases} \quad (3)$$

$$\begin{cases} \frac{dy_{ra}}{dt} = Q_5 i_{sa} - Q_6 y_{ra} - \omega_r y_{rb} \\ \frac{dy_{rb}}{dt} = Q_5 i_{sb} - Q_6 y_{rb} - \omega_r y_{ra} \end{cases} \quad (4)$$

Where: $Q_1 = \frac{R_s L_r^2 + R_r L_m^2}{L_s L_r^2 s}$; $Q_2 = \frac{R_s L_m}{L_s L_r^2 s}$; $Q_3 = \frac{L_m}{L_s L_r s}$; $Q_4 = \frac{1}{L_s s}$; $Q_5 = \frac{R_r L_m}{L_r}$; $Q_6 = \frac{R_r}{L_r}$;

Differential equations (3) and (4) show the electromagnetic relationship between the current, rotor flux, and rotor speed quantities. Based on the features of these equations, precise control algorithms can be implemented to achieve the set requirements.

Field-oriented control technique for the induction motor drive

Besides the advantages of stability, durability, and economy, a three-phase induction motor still has two disadvantages in the traditional operating control. There is always a slip between the reference speed and the motor's actual speed. The other is the interaction between magnetic flux and torque during the control process (the non-independence of magnetic flux and torque control). The FOC method is an effective technique for overcoming these limitations in the speed control of the induction motor. In the FOC method, the stator current vector is decomposed into two orthogonal components in the rotating reference frame [x-y]. The component i_{sx} , aligned with the rotor flux vector, is responsible for controlling the flux amplitude, while the perpendicular component i_{sy} controls the torque and thus the motor speed^{7,12}. This separation allows independent control of flux and torque, similar to the principle of a separately excited DC motor.

Fig. 1 presents a block diagram of the modern induction motor drive (IMD) system, comprising the following main components: a controller consisting of a computer and a Digital Signal Controller, a voltage-source inverter, and a motor (connected to the load). The control scheme for the IMD system based on the FOC technique is described in detail in Fig. 2.

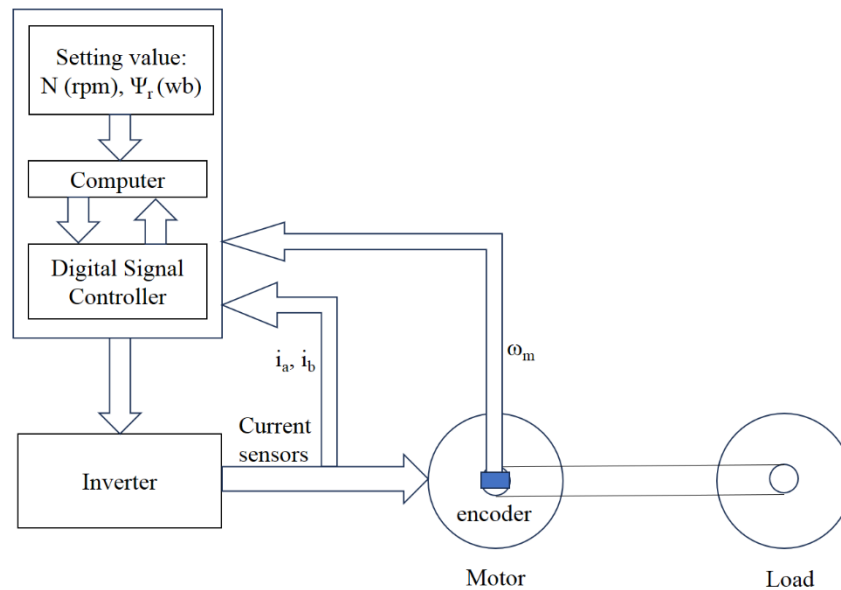


Fig. 1. Block diagram of an induction motor drive.

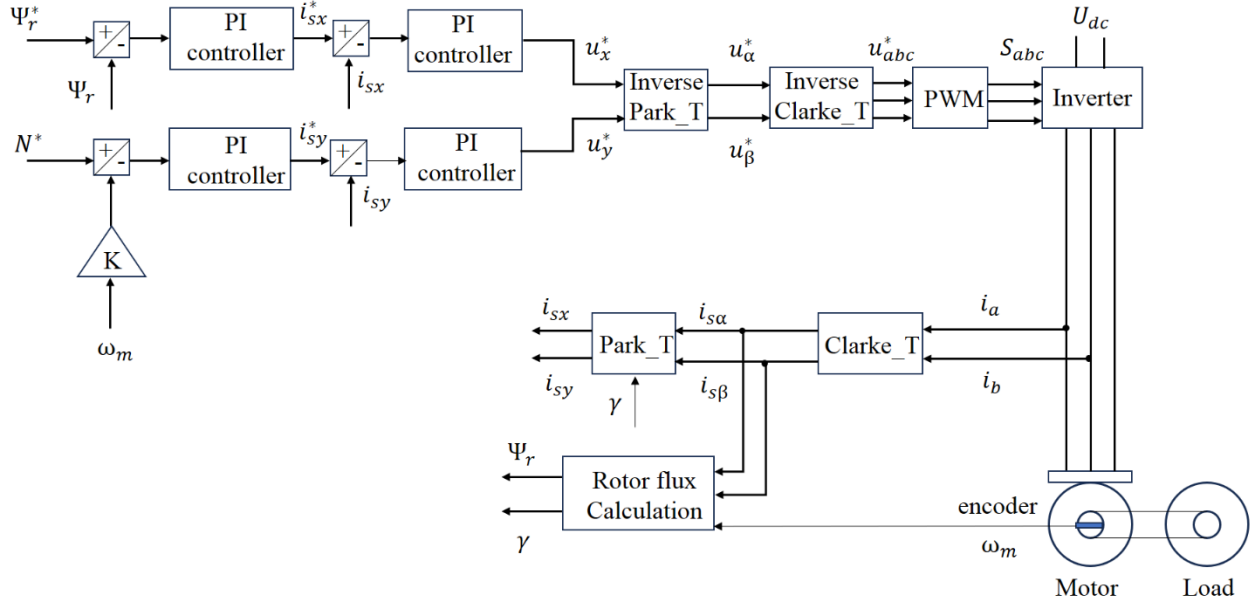


Fig. 2. FOC structure for induction motor drive.

Speed control using the FOC strategy can be summarized as follows:

- First step: Set the reference value: demand motor speed, nominal rotor flux,
- Second step: Operate in the initial condition,
- Third step: The phase currents from current sensors and rotor speed from the encoder are feedback to the FOC controller,
- In the FOC controller:
 - o The phase current in [a-b-c] coordinate is transformed into the stationary [α - β] coordinate by the Clarke transform;
 - o Stator current in [α - β] coordinate is combined with feedback rotor speed to calculate the rotor flux elements, such as amplitude and angle, by applying the current model or the voltage model^{7,12};
 - o The current space vector in [α - β] coordinate is combined with the rotor flux angle to transfer into [x-y] coordinate by Park transform.
- Fourth step: The calculated quantities are combined with the setting values to generate reference voltages by the typical PI controllers,
- Fifth step: The reference voltage is modulated into switching pulses that control the inverter to supply power to the motor,
- Repeat: Implement the control loop in the third step.

Fault-tolerant control based on rotor flux observers

A sensor-fault protection function is integrated into the FOC controller to enhance the stability and reliability of drive system operation, as shown in Fig. 3.

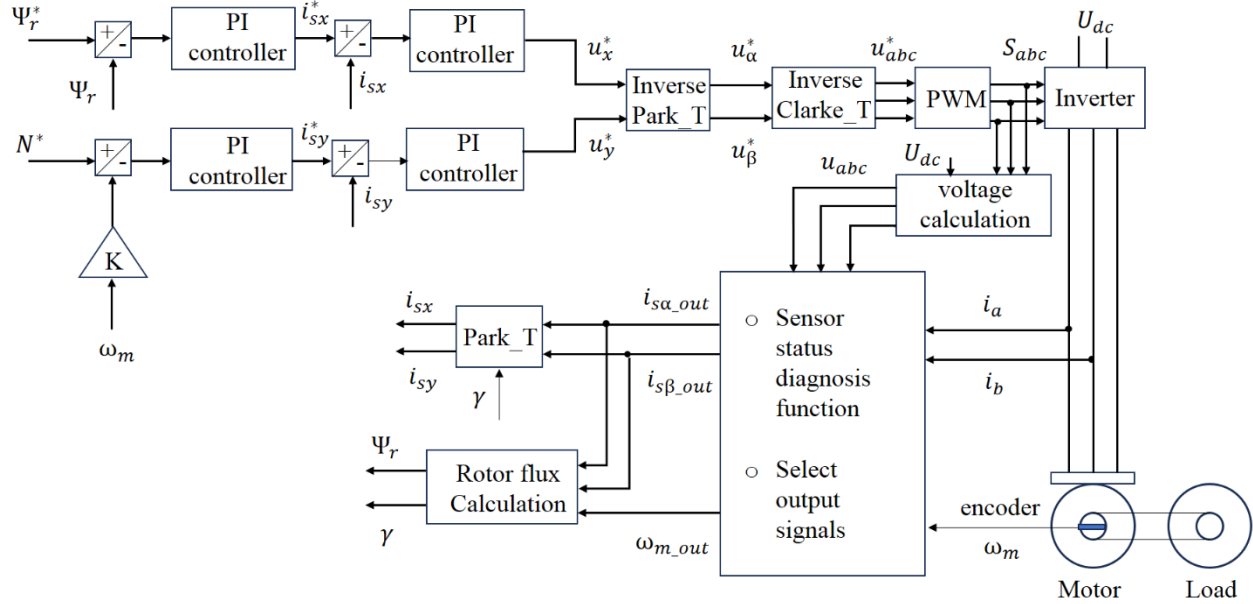


Fig. 3. IMD based on the FOC method integrates the FTC function.

Upon receiving signals from the sensor, the FTC function first diagnoses the sensor's health status and then determines the appropriate current and speed signals to supply to the FOC controller. If the sensors are all working correctly and have good signal quality, the stator current and rotor speed components supplied to the controller will correspond to the measurement signals from the sensors; in this case, the control mode is full sensor mode. Conversely, suppose a current or speed sensor fails. In that case, the measured signal will be replaced by the corresponding estimated signal; the motor drive will operate in current sensorless or speed-sensorless mode, respectively.

This research proposes a method for diagnosing the quality of sensor signals based on the rotor flux signals from two observers. The mathematical model of the two observers is presented in the following sections:

Types of typical sensor faults

During operation, especially in harsh environments, solid particles or liquid contaminants, as well as impact, shock, or vibration, may damage the speed sensor. These factors may lead to mechanical damage or output signal failure of the speed sensor (loose connection, disconnection failure, pulse failure, etc.)^{37,38}. In the case of current sensors, long-term operation can gradually degrade their performance, resulting in poor signal quality. Some typical faults of current sensors are described in Table 2³⁹:

Type faults	Mathematical expression	Note	
Speed sensor failures			
Partial fault of individual pulses	$\omega_{enc}=(1-\varsigma)\omega_m$	$-1<\varsigma<1$	$\square$ ω_{enc} : measured speed
Cyclic interruption of specific pulses		$\varsigma=[0,1]$	
Total fault		$\varsigma=1$	$\square$ ω_m : real speed $\square$ ς : constant-coefficient
Current sensor failures			
Soft fault: Drift-fault	$i_{meas}(t)=i(t)+a+bt$	$\square$ $i_{meas}(t)$: measured current; $\square$ $i(t)$: the true value;	
Soft fault: Scale-fault	$i_{meas}(t)=Ki(t)$		
Soft fault: Bias-fault	$i_{meas}(t)=i(t)+C$		

Soft fault: Precision degradation-fault	Precision	$i_{\text{meas}}(t) = i(t) + \text{err}(t)$	$\square$ a, b, C, K: constants corresponding to the fault types; $\square$ err(t): excessive random noise; $\square$ t: time variable.
Hard fault: Constant-fault		$i_{\text{meas}}(t) = C$	
Hard fault: Constant with noise-fault		$i_{\text{meas}}(t) = C + \text{err}(t)$	
Hard fault: Bottom noise-fault		$i_{\text{meas}}(t) = 0 + \text{err}(t)$	

Table 2. Expressions of the sensor fault types.

Proposed sensor fault diagnosis strategy based on rotor flux observers

In the FOC framework, the rotor flux is maintained at the rated value throughout the control process. Based on this basic idea, the proposed method uses rotor flux magnitudes from different estimators for sensor fault diagnosis. The voltage model (VM) is constructed based on the characteristic equation of the electrical machine model in the stationary coordinate system [α - β]; the voltage model uses voltage and current as input signals to observe the rotor flux vector during operation. Equations in (5) detail the method of estimating the rotor flux component of the VM:

$$\begin{aligned}
 \text{Voltage model: } \begin{cases} \frac{dy_{ra}^{VM}}{dt} = \frac{L_r}{L_m} u_{sa} - \frac{R_s L_r}{L_m} i_{sa} - \frac{\sigma L_s L_r - L_m^2}{\zeta} \frac{d^2 i_{sa}}{dt^2} \\ \frac{dy_{rb}^{VM}}{dt} = \frac{L_r}{L_m} u_{sb} - \frac{R_s L_r}{L_m} i_{sb} - \frac{\sigma L_s L_r - L_m^2}{\zeta} \frac{d^2 i_{sb}}{dt^2} \\ y_r^{VM} = \sqrt{(y_{ra}^{VM})^2 + (y_{rb}^{VM})^2} \end{cases} \quad (5)
 \end{aligned}$$

The current model (CM) uses the rotor current and speed as input signals to observe the corresponding rotor flux vector. Equations in (6) detail the method for estimating the rotor flux component of the CM:

$$\begin{aligned}
 \text{Current model: } \begin{cases} \frac{dy_{ra}^{CM}}{dt} = \frac{L_m}{T_r} i_{sa} - \frac{1}{T_r} y_{ra}^{CM} - w_r y_{rb}^{CM} \\ \frac{dy_{rb}^{CM}}{dt} = \frac{L_m}{T_r} i_{sb} - \frac{1}{T_r} y_{rb}^{CM} + w_r y_{ra}^{CM} \\ y_r^{CM} = \sqrt{(y_{ra}^{CM})^2 + (y_{rb}^{CM})^2} \end{cases} \quad (6)
 \end{aligned}$$

Two rotor flux observers, including a VM have been shown to maintain stability over the entire operating range, including the regeneration mode, and CM is stable by adapting the phase shift adaptation rule^{40,41}.

Besides the two rotor flux observers, the sensor fault diagnosis algorithm uses estimated speed and current signals. Any high-performance sensorless method can be applied to the proposed FTC technique, including RF-MRAS, CB-MRAS, SMO, etc., for rotor speed estimation³¹⁻³⁴, or Luenberger observer, direct calculation method, etc., for stator current estimation^{35,36}.

Remark: Depending on the sensorless control technique used, the computational efficiency of the proposed method can vary. The proposed method, when applied to the RF_MRAS technique in speed-sensorless mode, can simultaneously use both the voltage and current models for diagnosis and reconfiguration. This combination structure can support the proposed method to reduce the computational burden to suit applications with low-cost hardware. In other sensorless techniques, if a voltage or current model is included in the estimation algorithm, it can serve as a part of the diagnosis algorithm to lower computational costs.

The FTC function integrated into the controller is briefly described in the block diagram in Fig. 4, which includes the following processes: receiving sensor signals as input, evaluating sensor health status, and outputting warning flags for each sensor. Based on the health status of the sensors, the proper current and speed signals will be selected to transfer to the FOC controller for speed control.

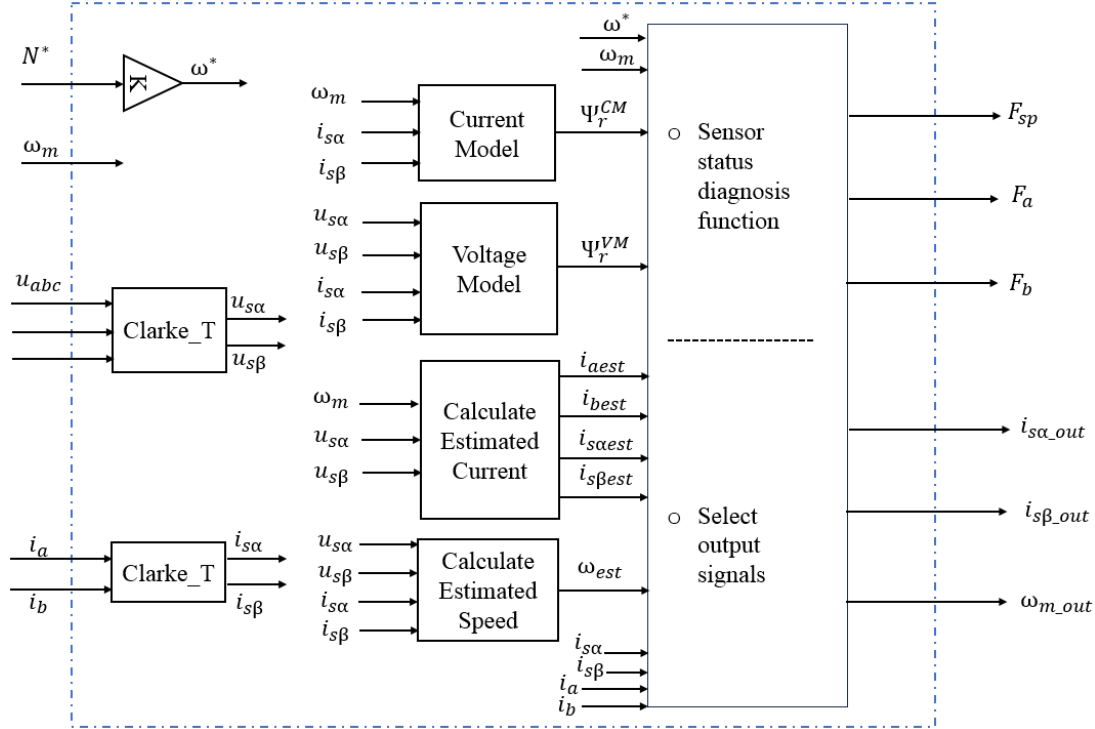


Fig. 4. Block diagram of FTC function in the FOC controller.

The proposed sensor fault diagnosis function uses two observers to evaluate the rotor flux and determine whether a sensor signal fault has occurred. Based on the deviation between the measured and estimated signals, it is possible to accurately identify which sensor is faulty.

Step 1: The rotor fluxes of the two observers are compared for similarities, as shown in equation (7); this step helps determine whether a signal failure has occurred.

$$\begin{aligned} \text{Index}_{\text{Fpsi}} &= |y_r^{\text{VM}} - y_r^{\text{CM}}| \\ \text{If}(\text{Index}_{\text{Fpsi}} > Th_{\text{psi}}) \{F_{\text{psi}} = 1;\} \\ \text{else} \{F_{\text{psi}} = 0;\} \end{aligned} \quad (7)$$

Step 2: A complex function is proposed to accurately determine the speed sensor's error status based on the Fpsi flag and a similarity comparison of the reference, measured, and estimated speeds. Equations in (8) are used to assess the speed sensor's failure in advance. If the measured speed signal is faulty, the Fsp flag will increase to 1.

$$\begin{aligned} \text{Index}_{\text{wrefsen}} &= |w^* - w_{\text{sen}}|; \\ \text{Index}_{\text{westsen}} &= |w_{\text{est}} - w_{\text{sen}}|; \\ \text{If}(F_{\text{psi}} \& \& (\text{Index}_{\text{wrefsen}} > Th_w) \& \& (\text{Index}_{\text{westsen}} > Th_w)) \{F_{\text{sp}} = 1;\} \\ \text{else} \{F_{\text{sp}} = 0;\} \end{aligned} \quad (8)$$

Step 3: Equations (9) and (10), which compare each phase current between the measured and estimated signals, can identify which current sensor is faulty. If the measured phase-A

current signal is defective, the F_a flag will increase to 1 in equation (9); if the measured phase-B current signal is defective, the F_b flag will increase to 1 in equation (10).

$$\text{Index}_{ia} = |i_a - i_{a_est}|$$

$$\text{If}(\text{Index}_{ia} > Th_i) \{F_{cka} = 1;\} \text{else} \{F_{cka} = 0;\}$$

$$F_a = \overline{F_{sp}} \times F_{psi} \times F_{cka}$$

$$(9)$$

$$\text{Index}_{ib} = |i_b - i_{b_est}|$$

$$\text{If}(\text{Index}_{ib} > Th_i) \{F_{ckb} = 1;\} \text{else} \{F_{ckb} = 0;\}$$

$$F_b = \overline{F_{sp}} \times F_{psi} \times F_{ckb}$$

$$(10)$$

Th_ω and Th_i are the maximum angular rotor speed deviation and stator currents deviation under normal operating conditions. According to previous studies and simulation tests, a threshold of 5% to 10% of the rated value is appropriate for diagnosing sensor faults (a threshold of 10% is recommended for common types of sensor faults). A flowchart for the sensor fault diagnosis algorithm is shown in Fig. 5.

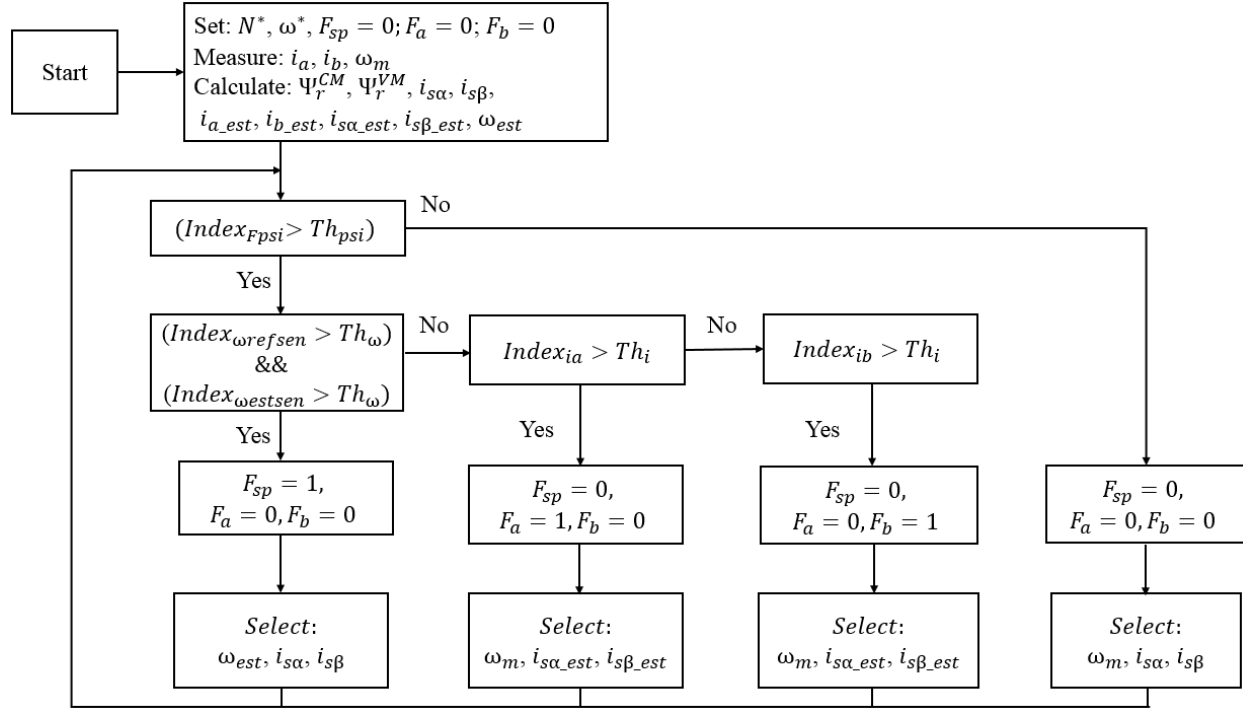


Fig. 5. Flow chart of sensor fault diagnosis strategy based on rotor flux observers.

The IMD integrates the proposed FTC technique to diagnose the health status of each sensor through the quality of the measured signal. If the measured signal is of good quality, the IMD should perform at its best across the entire sensor mode. If the speed sensor fails, the speed-sensorless mode should be used; alternatively, the current sensorless mode can be applied in case of current sensor failure. The IMD can maintain stable operation even under sensor failure conditions. The failure status can be identified explicitly through the corresponding flags. The sensors can be repaired or replaced at the appropriate time without affecting the economic efficiency and production process. Table 3 summarizes the health status of the sensors corresponding to the indicator flags and output signals provided to the FOC speed controller.

Sensor status	Indicator flags	Output signals
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Healthy	$Fa=0$, $Fb=0$, $Fsp=0$	i_{sa} $i_{s\beta}$ ω_m
Speed sensor fault	$Fa=0$, $Fb=0$, $Fsp=1$	i_{sa} $i_{s\beta}$ ω_{est}
A-phase current sensor fault	$Fa=1$, $Fb=0$, $Fsp=0$	i_{sa_est} $i_{s\beta_est}$ ω_m
B-phase current sensor fault	$Fa=0$, $Fb=1$, $Fsp=0$	i_{sa_est} $i_{s\beta_est}$ ω_m

Table 3. Status, sensor indicator flag, and FTC function output signals.

Simulation results and analysis

To comprehensively evaluate the proposed FTC method, various sensor fault types are simulated during the IMD's operation to assess its accuracy in detecting faulty sensors.

In evaluating the simulation cases, the IMD employs the FOC method, which combines the FTC function for speed control. The RF-MRAS technique³⁰ and the Luenberger observer³⁵ have been selected to generate estimated speed and stator currents. Three sensor fault groups are simulated to test the effectiveness of the proposed diagnostic method, including the speed sensor fault group, the soft fault group for the A-phase current sensor, and the hard fault group for the B-phase current sensor.

Starting with the evaluations, the IMD system operates at a reference speed of 500 rpm under a load of 7.4 Nm in full sensor mode, with all sensors in good condition. Fig. 6(a) shows that the measurement signals, which include two-phase current and feedback rotor speed, are stable and of high quality. Fig. 6(b) indicates that the sensor fault indicators are at low levels, corresponding to zero values. Fig. 6(c) illustrates that the motor speed matches the reference speed.

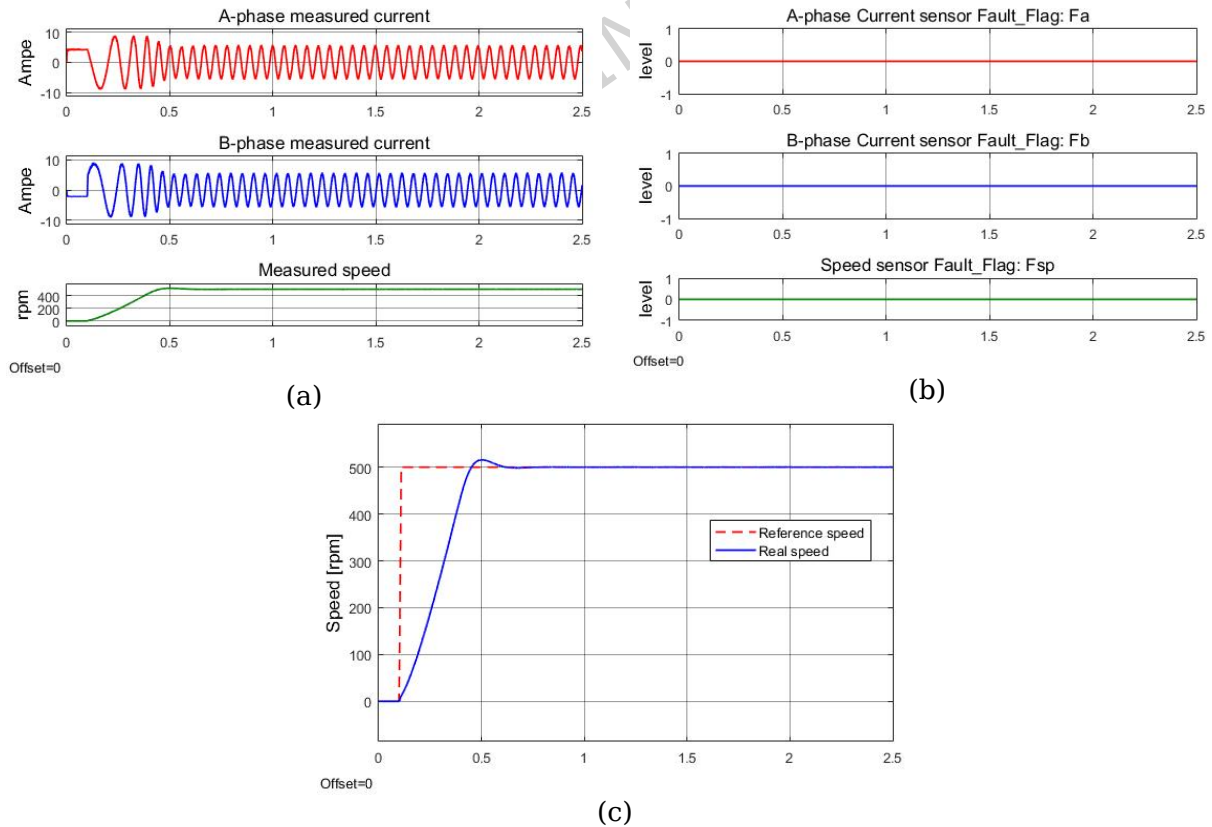
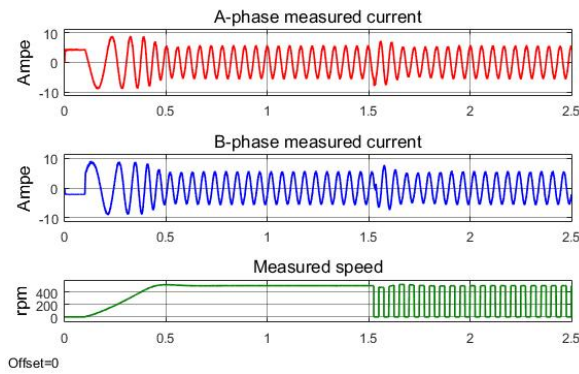
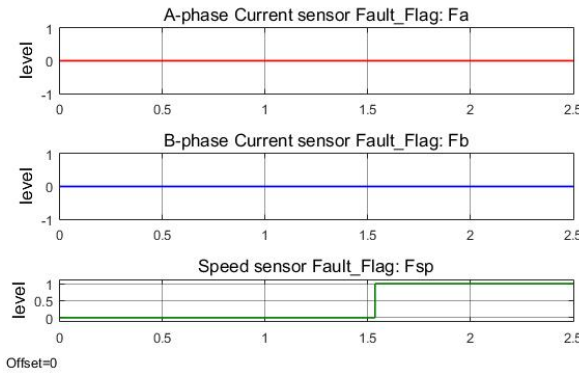


Fig. 6. Performance of the proposed FTC method under normal conditions: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.

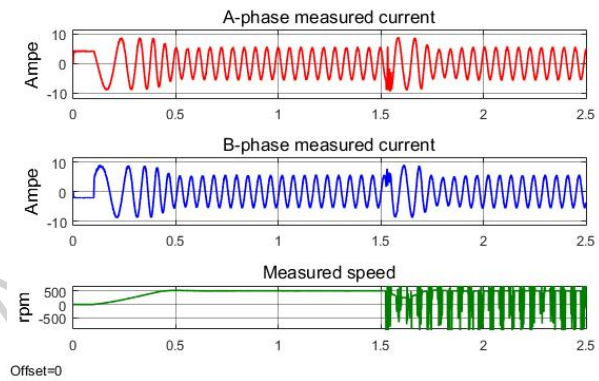
Each sensor's failure, as listed in Table 2, is simulated. Speed sensor faults in three cases are also simulated, and the effectiveness of the proposed FTC technique is examined in detail. The IMD system is operating stably. At 1.5 seconds, the speed sensor fails, corresponding to the failure of individual pulses (Fig. 7(a)), the cyclic interruption of specific pulses (Fig. 8(a)), or a total fault (Fig. 9(a)). In all three fault cases, the proposed FTC method works effectively, correctly identifying the faulty speed sensor. The F_{sp} flag rises to level 1, while the two flags, F_a and F_b , remain at level 0 in Fig. 7(b), Fig. 8(b), and Fig. 9(b). The faulty signal is isolated immediately after the corresponding fault flag is raised to level 1, and the estimated speed is used to replace the measured speed supplied to the FOC controller of the IMD system. The actual speed of the motor fluctuates for a short period and then quickly returns to stability, as shown in Figs. 7(c), 8(c), and 9(c).



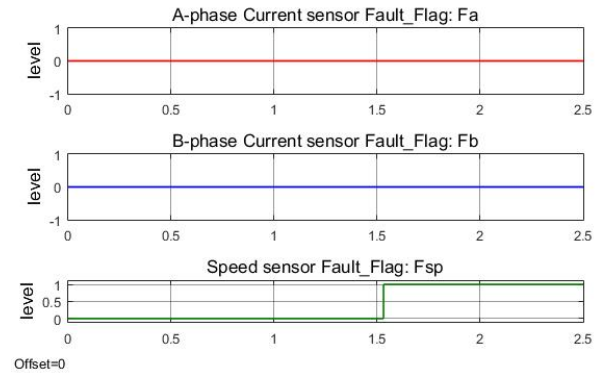
(a)



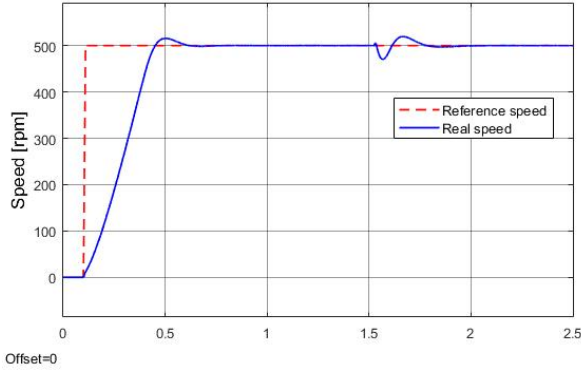
(b)



(a)

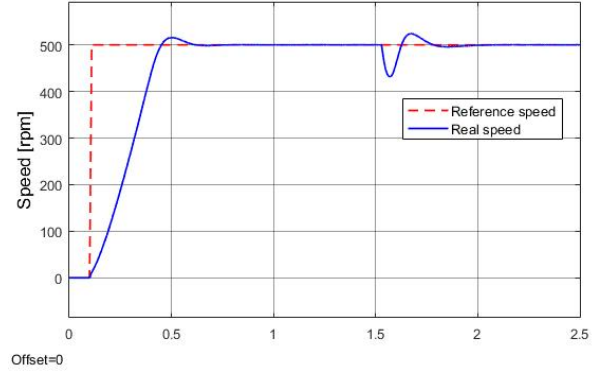


(b)



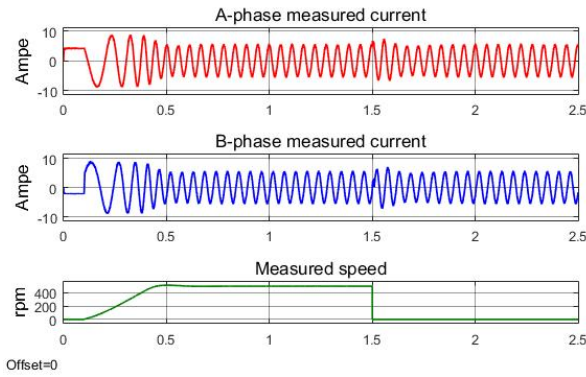
(c)

Fig. 7. Performance of the proposed FTC method under failure of individual pulses of speed sensor: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.

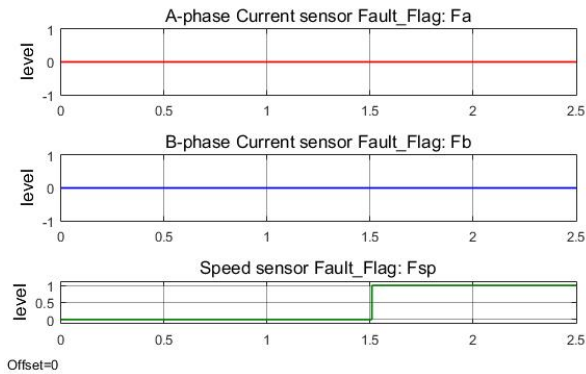


(c)

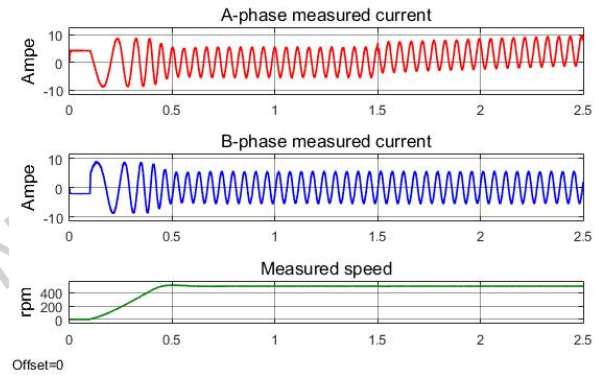
Fig. 8. Performance of the proposed FTC method under cyclic interruption of specific pulses from speed sensor: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.



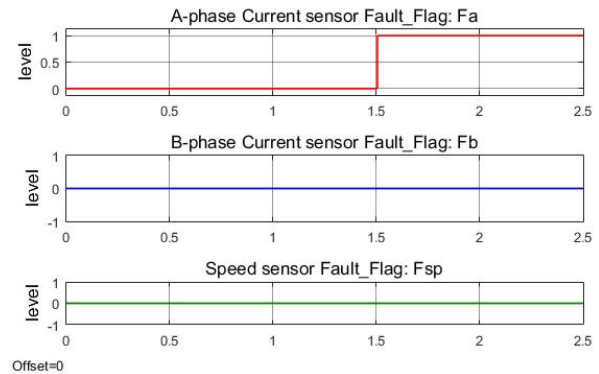
(a)



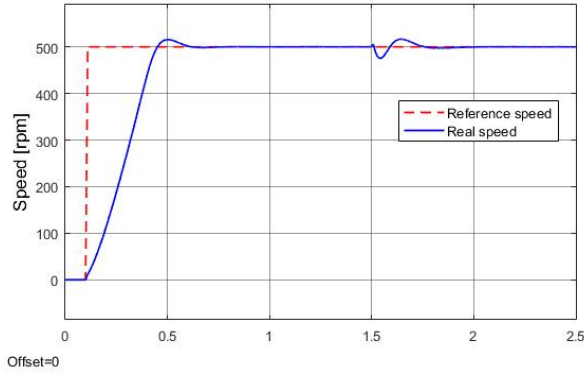
(b)



(a)

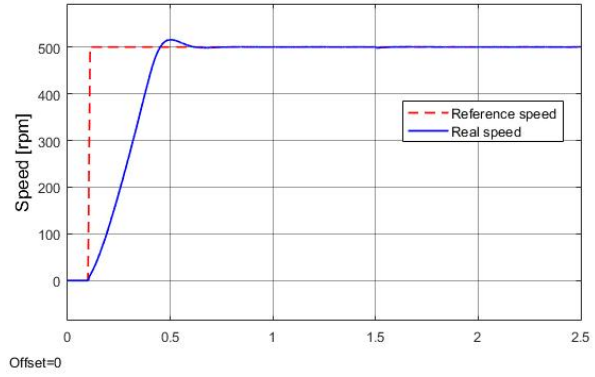


(b)



(c)

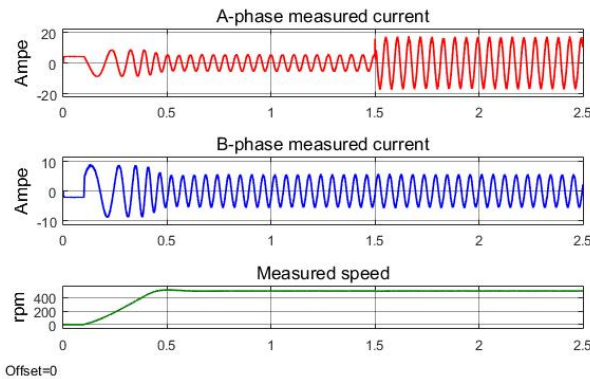
Fig. 9. Performance of the proposed FTC method under total fault of speed sensor: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.



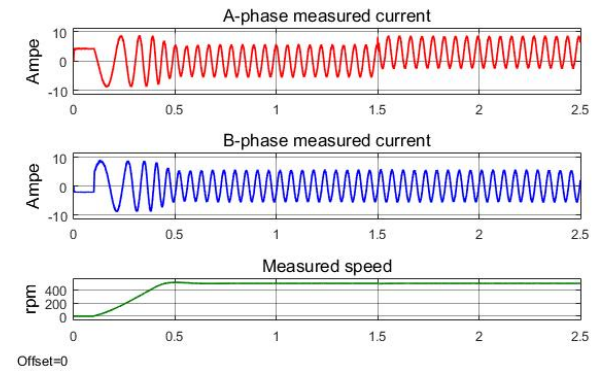
(c)

Fig. 10. Performance of the proposed FTC method under drift fault of A-phase current sensor: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.

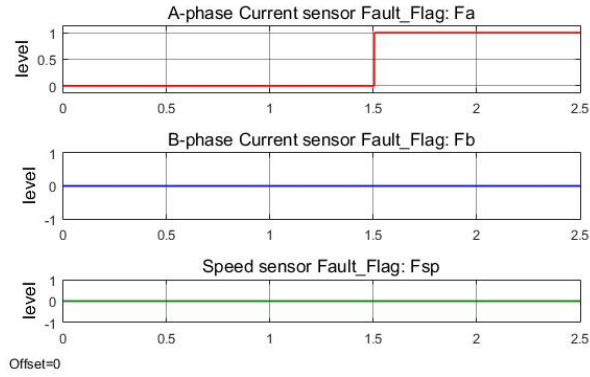
For current sensor faults, the soft fault group is executed for the A-phase current sensor, and the hard fault group is executed for the B-phase current sensor, respectively. The current sensor fault, including drift fault, scale fault, bias fault, and precision degradation fault, corresponds to Figs. 10(a), 11(a), 12(a), and 13(a), occurs with an A-phase sensor at 1.5 seconds. In all four failure cases, the FTC method worked effectively; the faulty sensor was correctly identified. The F_a flag rose to level 1, while the two flags, F_{sp} and F_b , remained at level 0, as shown in Figs. 10(b), 11(b), 12(b), and 13(b). The faulty current signal is isolated immediately, and the estimated current is used to replace the measured signal supplied to the FOC loop. The transient state occurs only briefly; the motor drive maintains stability effectively, as shown in Figs. 10(c), 11(c), 12(c), and 13(c). The hard fault group is verified to correspond to the B-phase current sensor. The total fault, constant noise, and bottom noise are assumed to occur with phase B, as shown in Figs. 14(a), 15(a), and 16(a). The FTC method works effectively; the F_b flag rises to level 1, while the two flags, F_{sp} and F_a , remain at level 0 in these three cases, as shown in Figs. 14(b), 15(b), and 16(b). Figs. 14(c), 15(c), and 16(c) have demonstrated the effectiveness of the proposed FTC; the IMD operated stably under the current sensor fault conditions.



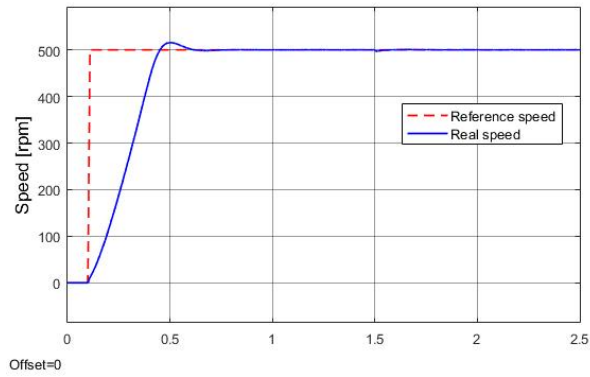
(a)



(a)

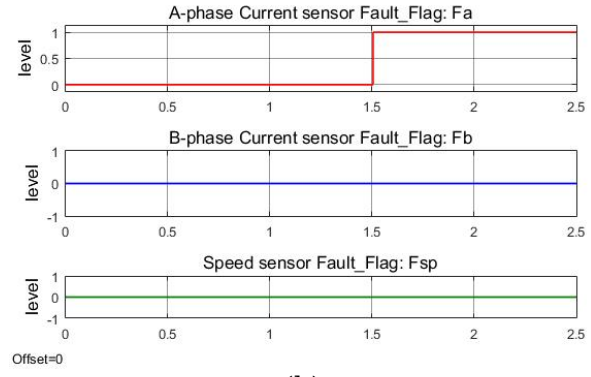


(b)

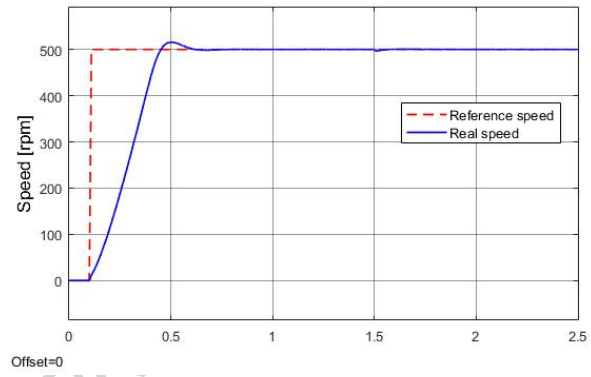


(c)

Fig. 11. Performance of the proposed FTC method under scale fault of A-phase current sensor: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.

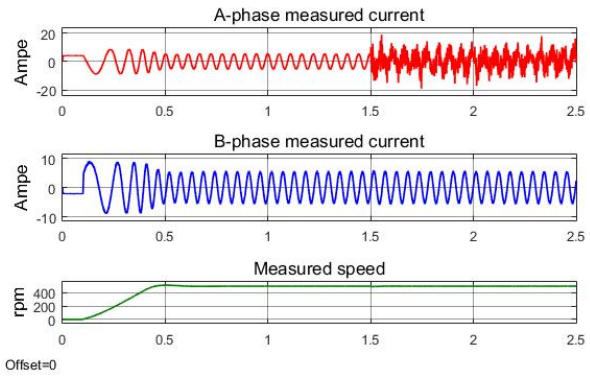


(b)

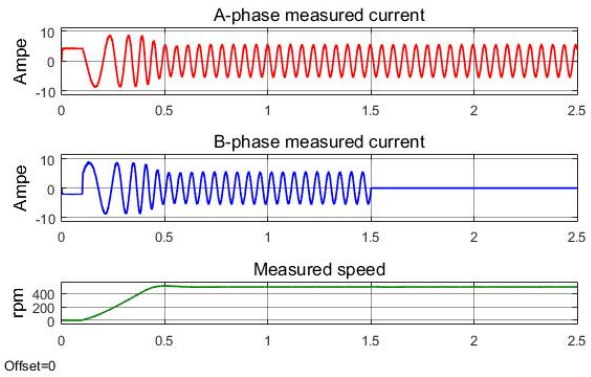


(c)

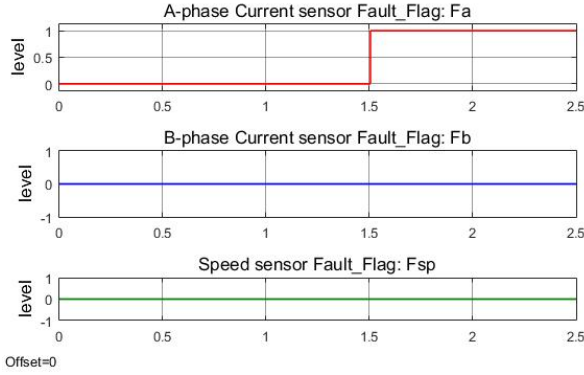
Fig. 12. Performance of the proposed FTC method under bias fault of A-phase current sensor: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.



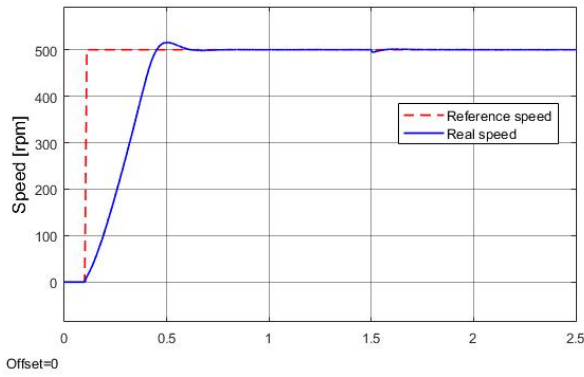
(a)



(a)

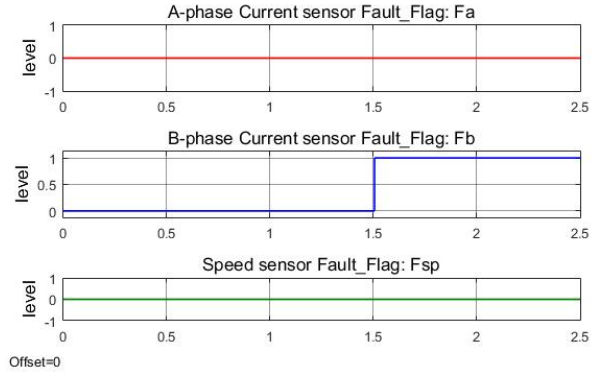


(b)

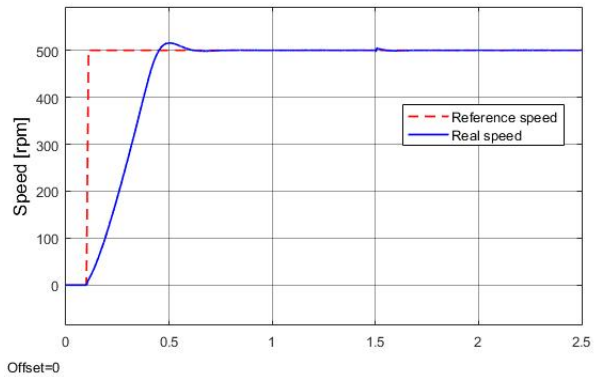


(c)

Fig. 13. Performance of the proposed FTC method under precision degradation-fault of A-phase current sensor: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.

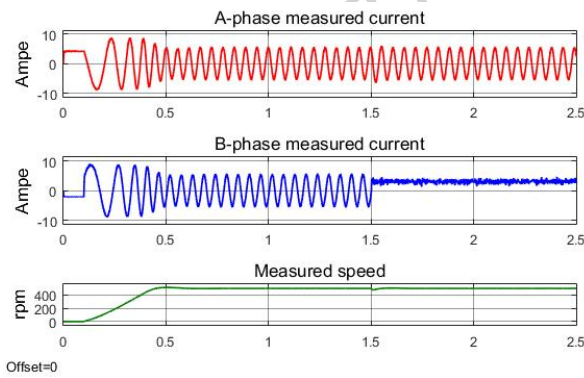


(b)

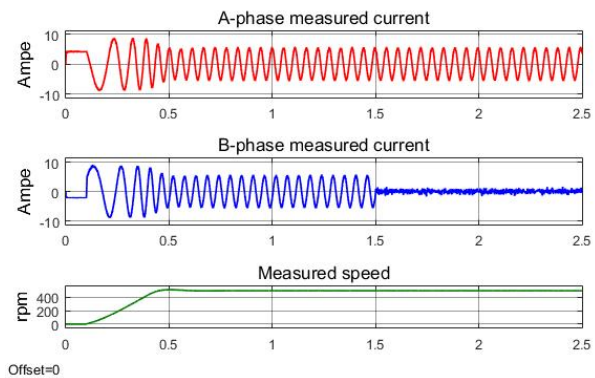


(c)

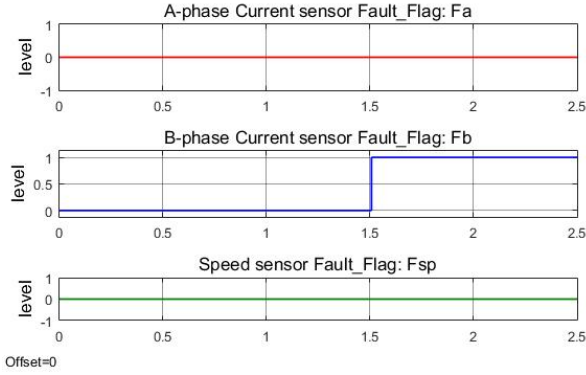
Fig. 14. Performance of the proposed FTC method under constant-fault (total fault) of B-phase current sensor: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.



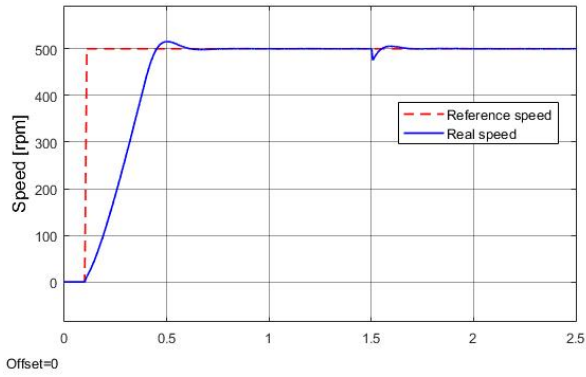
(a)



(b)

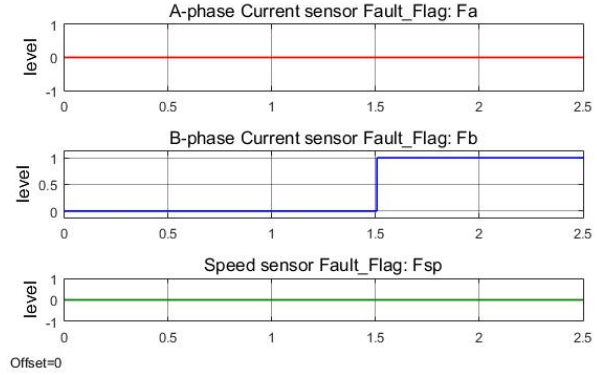


(b)

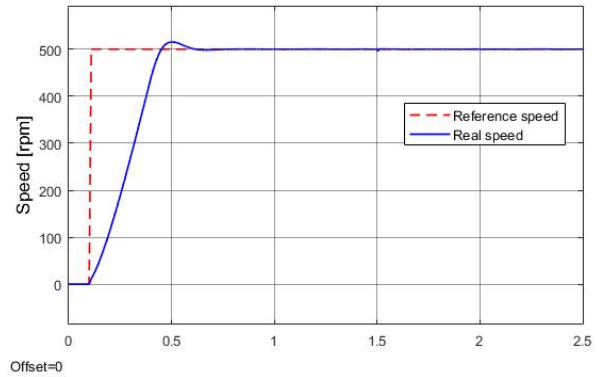


(c)

Fig. 15. Performance of the proposed FTC method under constant with noise-fault of B-phase current sensor: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.



(b)



(c)

Fig. 16. Performance of the proposed FTC method under bottom noise-fault of B-phase current sensor: (a)-Measured signal, (b)-Fault indication, (c)-Motor speed.

Through 10 simulated sensor fault cases, the FTC method demonstrated its effectiveness in accurately detecting faulty sensors, isolating and replacing control signals appropriately, ensuring that the IMD system maintains continuous operation under fault conditions, and providing specific health status indicators for each corresponding sensor, which supports the repair process at the right time without affecting continuous operation.

Experimental results and analysis

To validate the performance of the proposed fault-tolerant control (FTC) scheme in real-world operating conditions, a series of laboratory experiments was conducted using a fully instrumented induction motor drive (IMD) platform. The experimental setup was developed at the Intelligent Motor Control Laboratory and is schematically presented in Fig. 17.

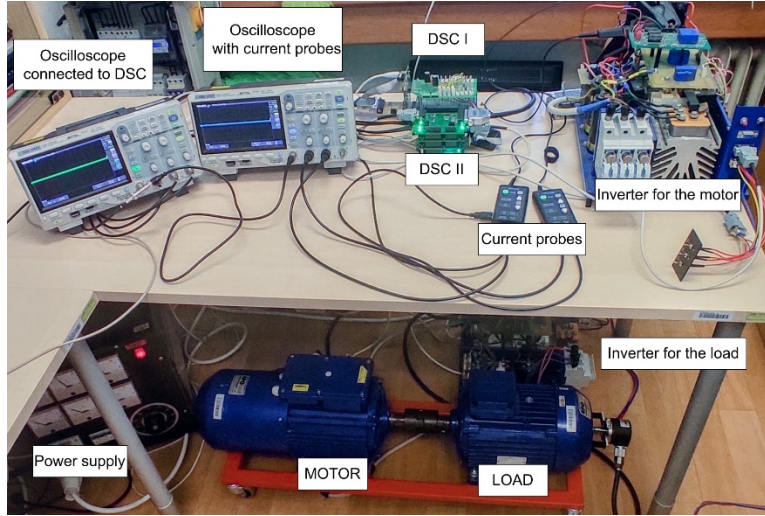


Fig. 17. Laboratory stand.

Hardware configuration

The test bench consists of a three-phase induction motor mechanically coupled to a controllable load machine. Independent voltage-source inverters power both the motor and the load. The motor-side inverter is controlled via a TMS320F28335 digital signal controller (Texas Instruments), while the system is supplied from a regulated three-phase power source. Real-time signal acquisition is performed using two oscilloscopes: one connected directly to the DSC to capture control feedback signals, and the other integrated with current probes to monitor stator phase currents. Two interface boards (DSC I and DSC II) were used to inject controlled sensor faults and capture diagnostic responses. All control algorithms, including flux observers and fault detection routines, were implemented in C++ within the Code Composer Studio (CCS) environment. The nominal parameters of the tested induction motor are summarized in Table 4.

Main electrical data	Value	Unit
Stator resistance	1.83	Ω
Rotor resistance	2.78	Ω
Stator inductance	0.319	H
Rotor inductance	0.318	H
Mutual inductance	0.309	H

Table 4. Nominal electrical parameters of the induction motor.

Experimental protocol and fault scenarios

1. Operates at 150 rpm with reverse direction in no-load mode

All experiments were conducted under constant operating conditions, with the motor speed set to 150 rpm. At $t = 4.0$ s, the rotation direction was reversed while maintaining the same speed level, and the system operated in no-load mode. The FTC system was evaluated under four distinct sensor fault conditions, each designed to emulate typical fault modes encountered in practical applications. Faults were introduced at $t > 3.0$ s, and data were recorded for control response, diagnostic flags, and motor behavior. The four investigated fault scenarios are presented in Figs. 18-21.

The first case is a complete loss of the speed sensor signal (Fig. 18). At the time of the fault, the rotor speed signal is completely disconnected. Fig. 18(a) shows that the measured signal is lost, while the FTC algorithm detects this deviation and triggers the speed sensor fault flag "Fsp" (Fig. 18(b)). Immediately afterwards, the estimated rotor speed is used to replace

the faulty measured signal. Fig. 18(c) shows that the actual speed quickly stabilizes, maintaining accurate control thanks to the FTC structure. In the second case, a speed sensor fault occurs, resulting in a scale fault, where a feedback signal twice the actual speed signal is sent back to the FOC controller, as shown in Fig. 19(a). The Fsp flag is triggered in time (Fig. 19(b)). After the fault signal is isolated, the estimated speed takes over the control role, helping the motor maintain stable operation as shown in Fig. 19(c).

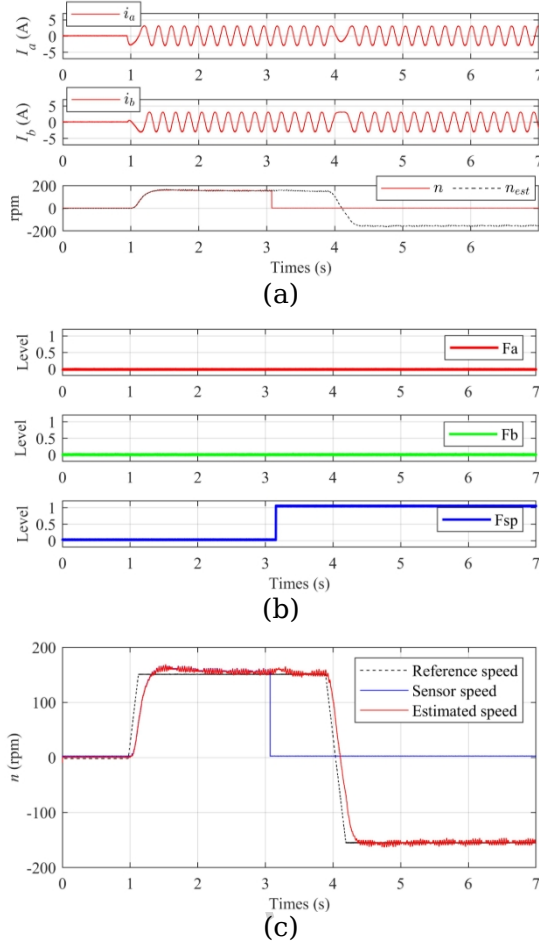


Fig. 18. Response of the proposed FTC method under total fault of the speed sensor: (a) Measured speed signal, (b) Speed fault flag Fsp, (c) Actual and estimated rotor speed.

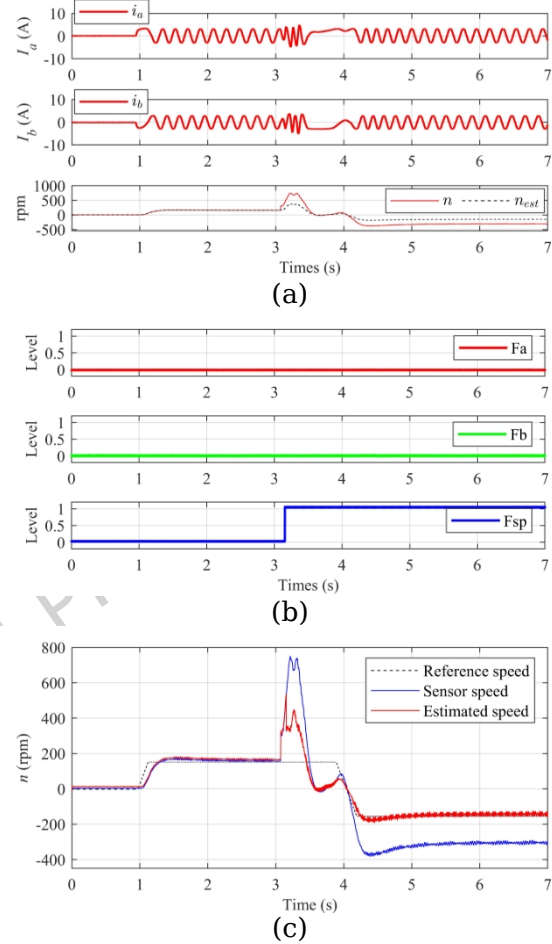


Fig. 19. Response of the proposed FTC method under double-speed fault of the speed sensor: (a) Measured speed signal, (b) Speed fault flag Fsp, (c) Actual and estimated rotor speed.

Case 3 involves a complete loss of the B-phase current signal (Fig. 20). The B-phase current sensor is entirely disconnected, resulting in the measured current dropping abruptly, as shown in Fig. 20(a), while the estimated current remains continuous. The fault flag "Fb" is correctly triggered, as illustrated in Fig. 20(b), and the estimated current is injected into the FOC loop to replace the faulty signal. Fig. 20(c) demonstrates that the motor speed remains stable, indicating the system's strong fault-tolerance capability. The final experimental case addresses a DC offset fault in the B-phase current sensor (Fig. 21). A fixed DC offset is added to the B-phase current measurement to emulate a constant bias. Fig. 21(a) shows a clear deviation between the measured and estimated signals. The FTC algorithm accurately identifies the fault and activates the fault flag "Fb", as shown in Fig. 21(b). After substituting

the faulty current with the estimated signal, Fig. 21(c) confirms that the motor speed is not significantly affected and that the system continues to operate stably.

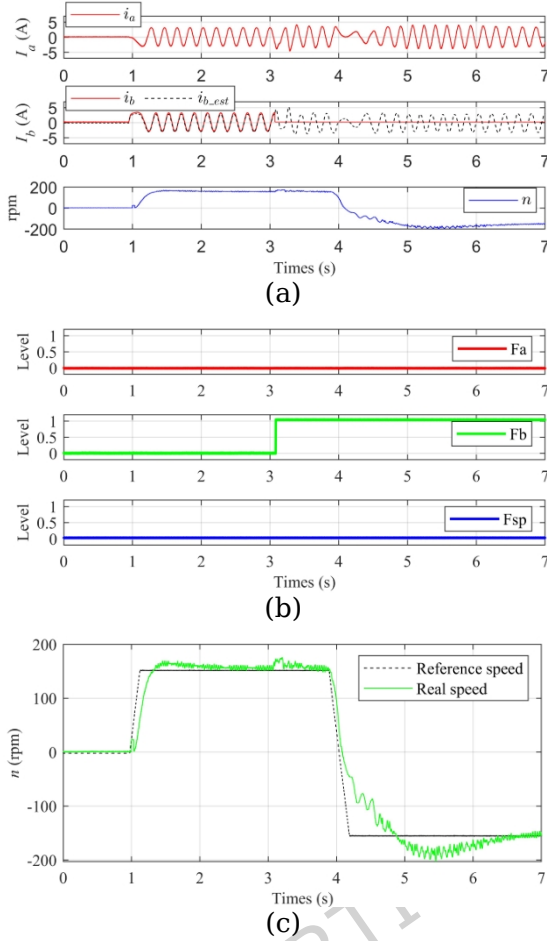


Fig. 20. Response of the proposed FTC method under total fault of the B-phase current sensor: (a) Measured and estimated current signals, (b) Current fault flag F_b , (c) Motor speed response.

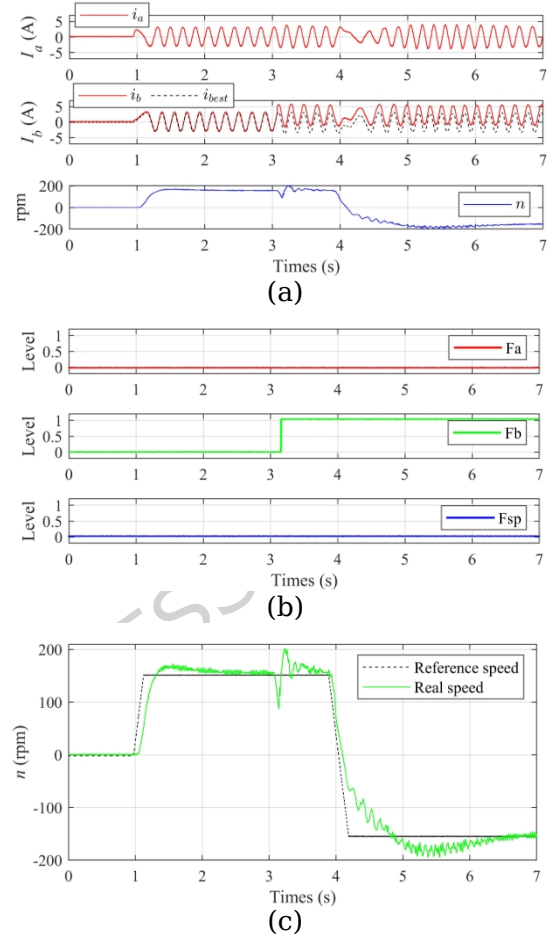


Fig. 21. Response of the proposed FTC method under offset fault of the B-phase current sensor: (a) Measured and estimated current signals, (b) Current fault flag F_b , (c) Motor speed response.

2. Variable-speed operation with direction reversal in no-load mode

In this set of experiments, the induction motor was operated in a variable-speed configuration under the same load conditions as in the above case. The speed was gradually increased from standstill to 300 rpm at $t = 0.5$ s, further increased to 400 rpm at $t = 2.2$ s, and then reversed starting at $t = 3.8$ s to reach -400 rpm at $t = 5.5$ s. This reversed speed was maintained until the end of the operating cycle.

Fig. 22 shows the response of the proposed FTC method in the case of a global speed sensor failure: In this case, the rotor speed feedback signal is completely interrupted during variable speed operation. As shown in Fig. 22(a), the measured speed signal drops to zero when the failure occurs. The FTC algorithm detects the deviation and triggers the speed fault flag F_{sp} (Fig. 22(b)). Immediately after that, the estimated rotor speed is used to replace the faulty measured signal. Fig. 22(c) shows that the actual speed closely follows the reference value during the acceleration, steady state, and reversal phases, confirming the effectiveness of the FTC.

Fig. 23 shows that the measured phase current A is fixed at a constant value of 3 A to simulate a severe sensor failure. Fig. 23(a) shows a clear deviation between the measured phase A current and the estimated current after the fault occurs. The FTC algorithm accurately identifies the anomaly and activates the Fa current fault flag (Fig. 23(b)). The estimated current in the FOC loop then replaces the faulty measurement. As shown in Fig. 23(c), the motor speed remains stable and accurately tracks the reference value throughout the operating cycle.

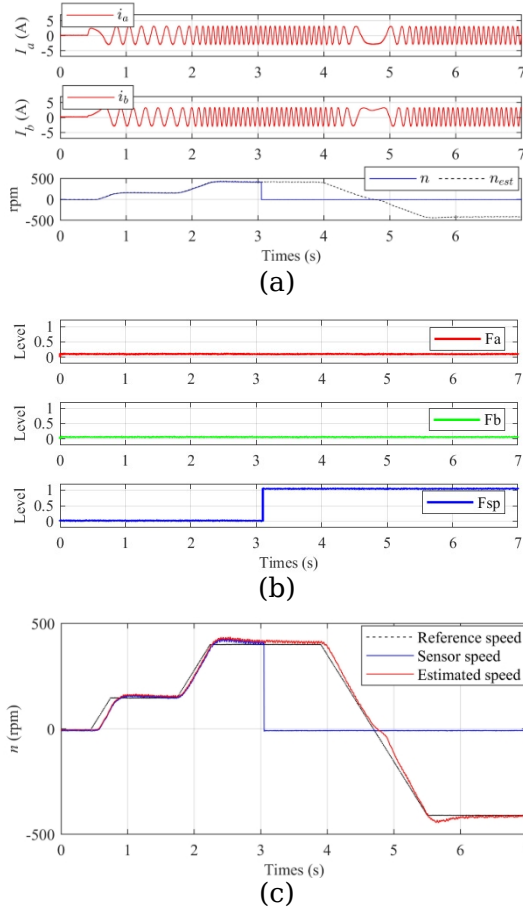


Fig. 22. Response of the proposed FTC method under total fault of the speed sensor: (a) Measured speed signal, (b) Speed fault flag Fsp, (c) Actual and estimated rotor speed.

In both cases in Figs. 24 and 25, the affected phase current (phase A in Fig. 24 and phase B in Fig. 25) drops abruptly to zero when the fault occurs, as shown in sub-figure (a). The estimated current remains valid, allowing the FTC algorithm to detect the anomaly quickly. The corresponding fault flags (Fa for phase A, Fb for phase B) are activated in sub-figure (b). Immediately, the faulty current measurement is replaced by the estimated current in the FOC control loop. As shown in sub-figure (c) in both figures, the rotor speed maintains stable operation and follows the reference precisely during the acceleration, steady state, and reversal phases. This demonstrates that the FTC scheme can effectively handle the total current sensing loss, regardless of which phase is affected.

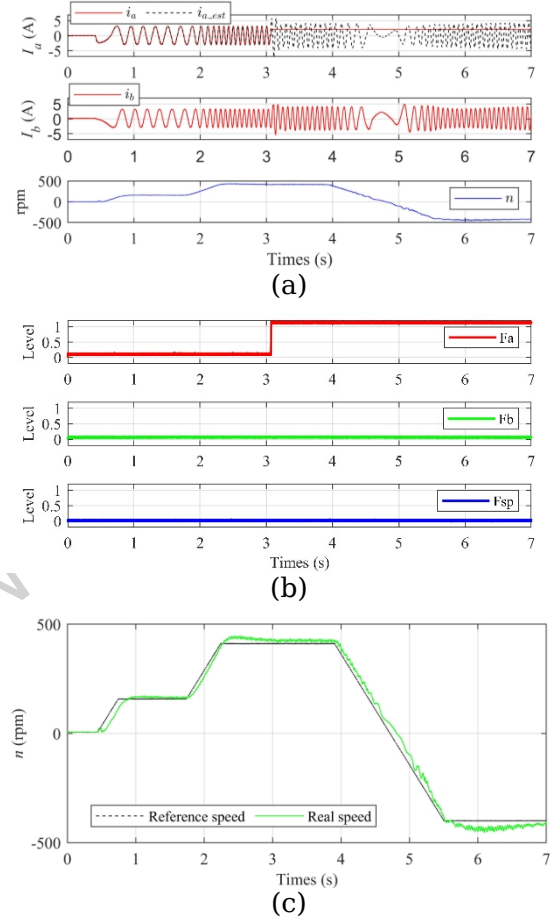


Fig. 23. Response of the proposed FTC method under constant fault of the A-phase current sensor: (a) Measured and estimated current signals, (b) Current fault flag Fa, (c) Motor speed response.

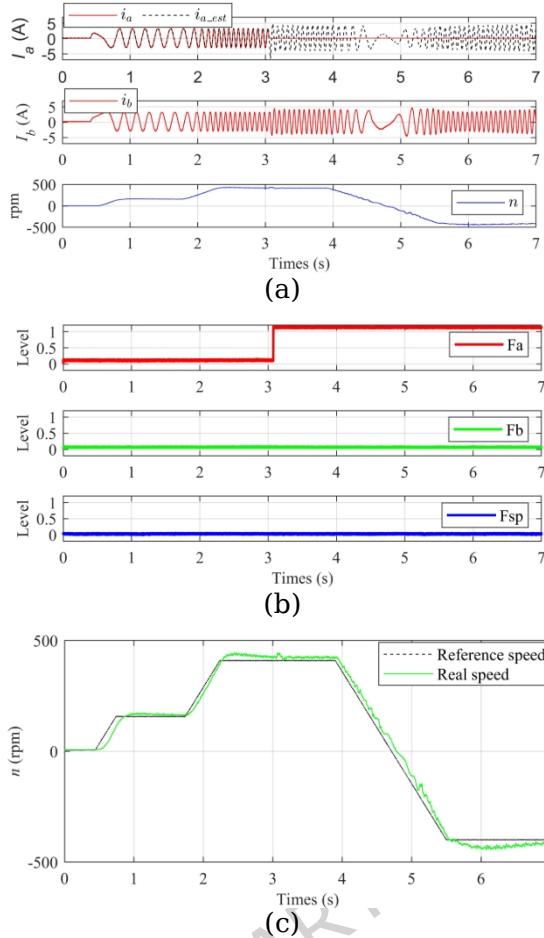


Fig. 24. Response of the proposed FTC method under total fault of the A-phase current sensor: (a) Measured and estimated current signals, (b) Current fault flag F_a , (c) Motor speed response.

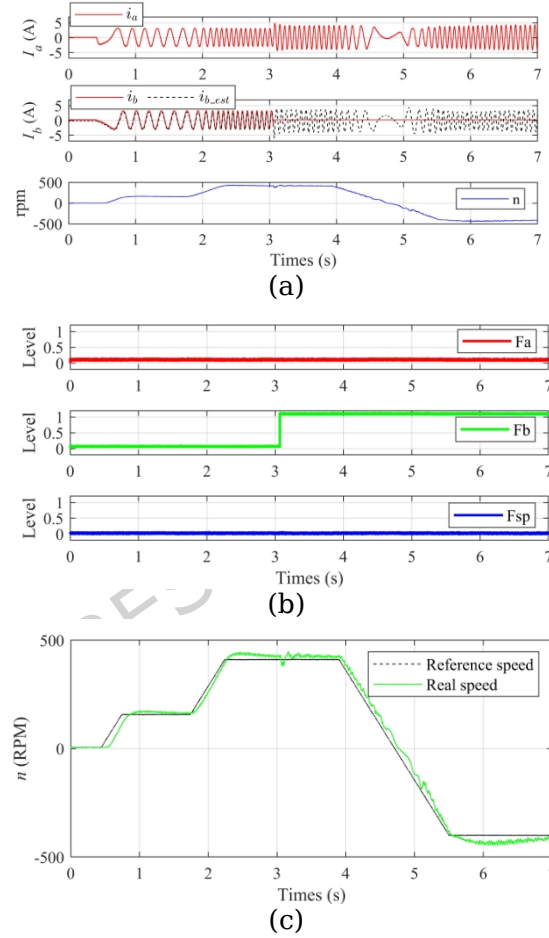


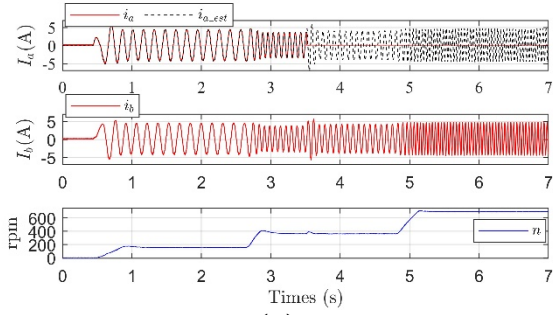
Fig. 25. Response of the proposed FTC method under total fault of the B-phase current sensor: (a) Measured and estimated current signals, (b) Current fault flag F_b , (c) Motor speed response.

3. Multi-Speed and Variable Load Operation

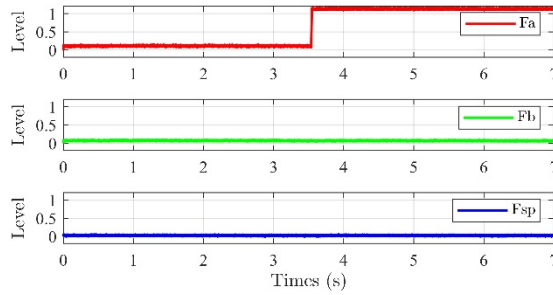
The motor accelerates from 0 rpm to 150 rpm at $t > 0.5$ s and operates stably until $t = 2.6$ s under full load (100% rated load). From $t = 2.6$ s to $t = 4.8$ s, the motor continues to accelerate to 350 rpm and carries 75% of the rated load. After $t = 4.8$ s, the motor continues to accelerate to 710 rpm while operating at 50% load.

In both cases shown in Figs. 26 and 27, the affected phase current (phase A in Fig. 28 and phase B in Fig. 29) suddenly drops to 0 when the fault occurs, as illustrated in sub-figure (a). The estimated current is still enabled, allowing the proposed FTC algorithm to quickly detect anomalies under different load conditions, where the current amplitude also changes accordingly. The corresponding faulty flags (F_a for phase A, F_b for phase B) are enabled in sub-figure (b). Immediately after the fault is detected, the faulty current measurement is replaced by the estimated current in the FOC control loop. As shown in sub-figure (c) of both figures, the rotor speed maintains stable operation and accurately tracks the reference speed

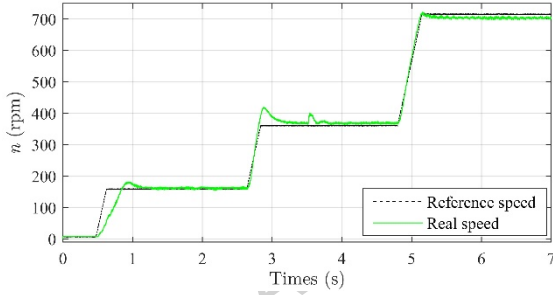
during the acceleration and steady-state phases. These results confirm that the proposed FTC scheme can effectively handle the overall current sensor faults, regardless of which phase is affected.



(a)

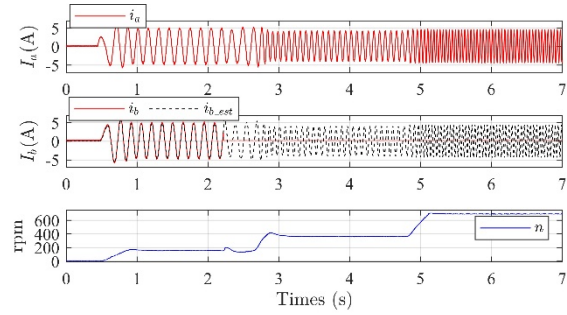


(b)

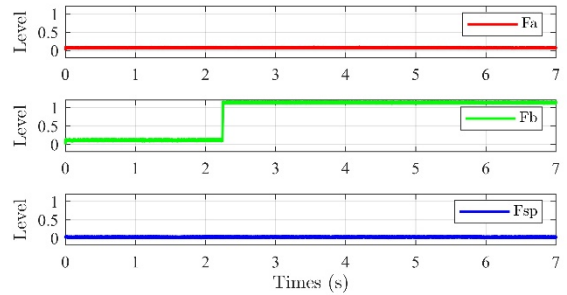


(c)

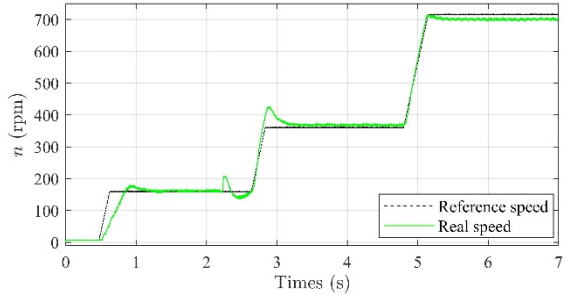
Fig. 26. Response of the proposed FTC method under total fault of the A-phase current sensor with variable load conditions: (a) Measured and estimated current signals, (b) Current fault flag F_b , (c) Motor speed response.



(a)



(b)



(c)

Fig. 27. Response of the proposed FTC method under total fault of the B-phase current sensor with variable load conditions: (a) Measured and estimated current signals, (b) Current fault flag F_b , (c) Motor speed response.

Fig. 28 illustrates the response of the proposed FTC method in the event of a complete speed sensor failure. In this case, the rotor speed feedback signal is completely interrupted during operation under variable speed and load conditions. As shown in Fig. 28(a), the measured speed signal suddenly drops to zero when the fault occurs, while the stator currents of phase A and phase B (i_a and i_b) remain stable and consistent with the current values under load. The proposed FTC algorithm promptly detects this abnormal deviation and triggers the speed fault flag F_{sp} , as shown in Fig. 28(b). Immediately after detection, the estimated rotor speed is substituted for the faulty measured signal in the control loop. Fig. 28(c) illustrates that the actual rotor speed accurately tracks the reference value during both the acceleration and steady-state phases, confirming the effectiveness and reliability of the proposed FTC scheme in the event of complete speed sensor loss.

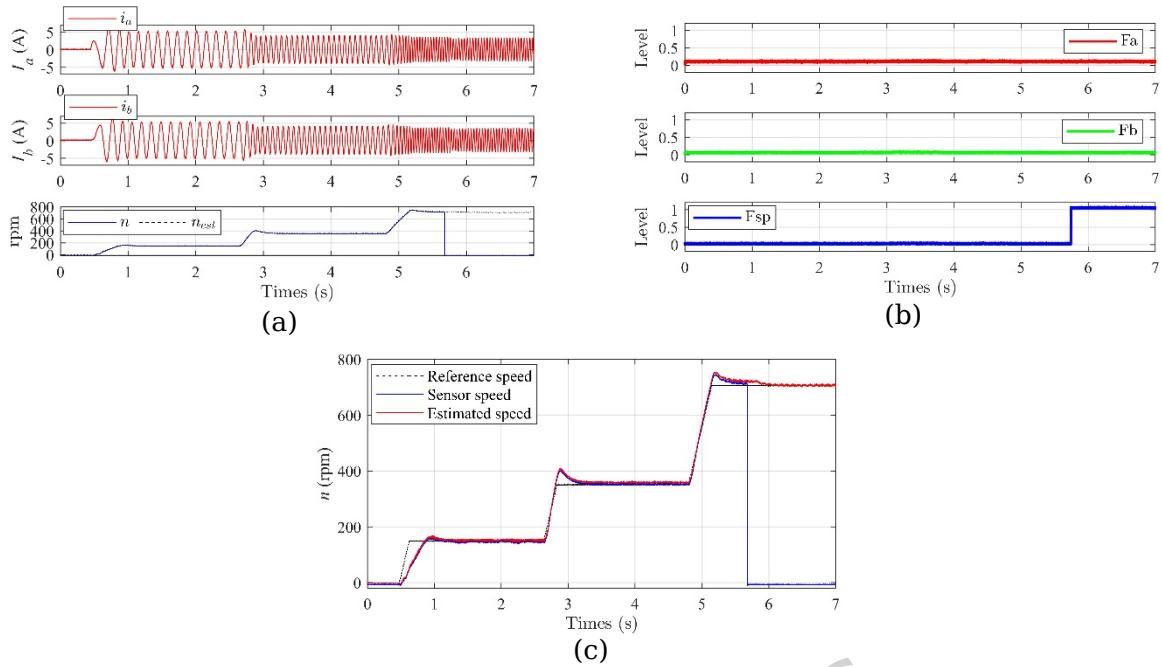


Fig. 28. Response of the proposed FTC method under total fault of the speed sensor with multi-stage speed and load conditions: (a) Measured speed signal, (b) Speed fault flag F_{sp} , (c) Actual and estimated rotor speed.

The experimental results demonstrate that the proposed FTC system is capable of detecting faults quickly and accurately, utilizing dedicated fault flags for each type of sensor. Once a fault occurs, the faulty measured signal is effectively isolated and replaced by a reliable estimated signal. The IMD continues to operate stably and accurately tracks the reference speed, even under serious fault conditions.

4. Accuracy Evaluation of the Proposed Fault Diagnosis Method under Sudden Reference Speed Changes

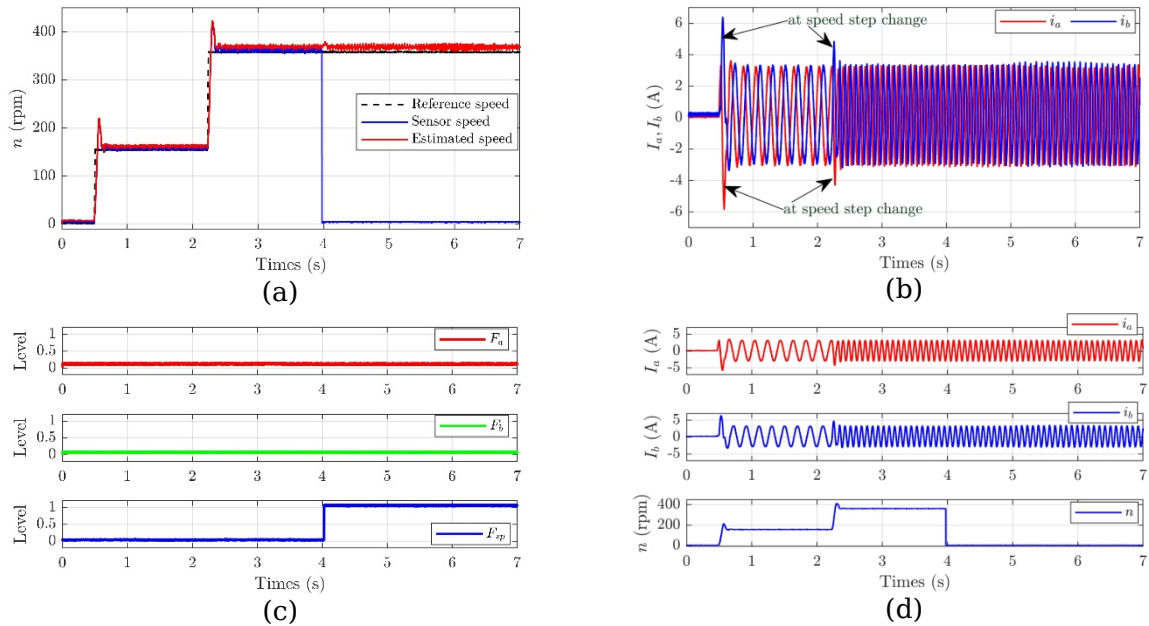


Fig. 29. Experimental results under sudden reference speed change using the proposed method: (a) Actual and estimated rotor speed, (b) Measured current signal, (c) Fault flag, (d) Measured signal.

To verify that the proposed fault diagnosis algorithm can accurately distinguish between real faults and transient variations, another experiment was conducted with a sudden change in the reference speed, as shown in Fig. 29. When the reference speed suddenly increases, the stator currents in phases A and B exhibit strong transient oscillations due to the instantaneous torque response of the induction motor, as illustrated in Fig. 29(b). Despite the rapid current fluctuations, the fault flag signals in Fig. 29(c) remain at the logic '0' level, indicating that the proposed algorithm correctly identified this as a normal transient phenomenon rather than a sensor fault. Furthermore, to evaluate the ability to withstand simultaneous disturbances, a speed sensor fault was generated at approximately $t = 4$ s. As shown in Fig. 29(d), the proposed algorithm accurately detected this fault. The estimated rotor speed in Fig. 29(a) closely followed the actual speed, confirming the effectiveness of the flux estimator in the proposed method.

These results demonstrate that the proposed approach not only avoids false alarms caused by sudden current fluctuations but also precisely identifies actual sensor faults under dynamic operating conditions. This verifies the diagnostic stability and reliability of the proposed method compared to traditional current observers.

Conclusion

This study presents a fault-tolerant control (FTC) method integrated into the field-oriented control loop to enhance the stability and reliability of an induction motor drive system (IMD) under sensor fault conditions. The proposed method utilizes rotor flux signals from the voltage model and current model to assess the consistency of sensor responses, employing a similarity check mechanism to accurately locate and classify faulty sensors. A series of diagnostic procedures enables the system to distinguish between typical speed sensor and current sensor fault types and reconfigure the control inputs accordingly using the estimated signals. Extensive simulations involving ten distinct fault scenarios demonstrate the method's ability to detect sensor faults with high accuracy. The FTC system ensures seamless switching between control modes, maintaining stable motor operation even in the event of sudden or severe faults. Fault flags were continuously assigned to the affected sensor, confirming the robustness of the diagnostic logic. To verify the practical applicability of the method, eleven fault cases with varying speeds were tested in a laboratory equipped with a TMS320F28335 controller under no-load and various load conditions. The experimental results closely matched the simulation results, indicating that the FTC system maintained accurate speed regulation and reliable fault isolation in all tested cases. The real-time diagnostic indicators also provided valuable information for timely sensor maintenance and replacement.

The distinctive feature of this method is to divide the diagnosis into two stages: one based on rotor-flux comparison (a quantity that varies less across different operating conditions) to determine whether a sensor signal fault has occurred, and the other to identify which sensor is faulty. This approach helps reduce misdiagnosis caused by faulty sensors. It is more adaptable for future research, enabling the development of FTC techniques across multiple diagnostic layers, with fallback options such as sensorless vector control and scalar control used in order of priority. However, the proposed fault diagnosis method has some limitations: it depends on the accuracy of stage 1 of the diagnosis function, which is crucial to overall accuracy. However, both rotor flux models are sensitive to variations in the motor's resistance parameters. Resistance estimation techniques can enhance accuracy, but they

will increase computational burden. Finding a balance between simplicity and computational cost efficiency is a challenge for future research.

Overall, the proposed FTC strategy provides a practical and effective solution to enhance fault resilience in IMD systems. Its modular design and real-time performance are particularly well-suited for deployment in industrial applications that require continuous and stable operation. Due to its simplicity and low computational burden, the proposed method can be applied to medium-performance processors if combined with suitable sensorless methods. Future studies will explore the integration of advanced sensorless techniques into this FTC framework to further improve performance under extended fault conditions and parameter uncertainties.

Data availability

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

C.D.T proposed the concept and methodology. C.D.T conducted simulations. M.K and V.S conducted the experiments. C.D.T, M.K, and P.D.N analyzed the results. All authors reviewed the manuscript.

Funding

This article was funded by the European Union under REFRESH, Research Excellence for REgion Sustainability and High-tech Industries, project number CZ.10.03.01/00/22_003/0000048, via the Operational Programme Just Transition and by the Student Grant Competition of VSB-Technical University of Ostrava, grant number SP2025/066.

Acknowledgements

The authors thank Ton Duc Thang University and VSB-Technical University of Ostrava for their support in this work.

Declarations

Competing interests

The authors declare no conflict of interest.

Additional information

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